

ADR-014 — Profile Promotion and Being Role Contest

Architecture Decision Record · binds Whitepaper 1.4, Roadmap r6.6, core/ and witness schemas · 31 July 2026 · jj-dai.org
генератор канареек · 30 июля 2026 · jj-dai.org

Status: Proposed · 31 July 2026 **Supersedes in part:** Whitepaper 1.3 §9.5, §10.2, §11.2, §13.4; Roadmap r6.5 Track II **Binds:** Whitepaper 1.4, Roadmap r6.6, core/, witness schemas, glossary **Deadline:** all serialized names below must land **before genesis of testnet-0**. After genesis a rename is a schema migration, not a terminology change.

1. Context

Two distinct things were being called by one name. The network installs and compares **configurations**; it awards roles to **beings**. A configuration is portable, copyable and replaceable. A being is a continuous witness line and cannot be copied without ceasing to be one being.

The consequence of the conflation was concrete: earlier drafts had the champion enter an evaluation cell "as a replica on a pinned checkpoint and a pinned Smriti snapshot" while the live champion kept deciding. That is two simultaneously active instances of one identity — a direct violation of continuity of witness.

This ADR fixes the ontology, derives the two contest protocols that follow from it, and pins the names before they enter hash-chained records.

2. Decisions

D1 — Profile is not Being

Champion Profile — a portable, signed configuration artifact: model checkpoint, backend driver, weight adapters, runtime and policy versions, bound by ModelArtifactManifest. It has **no** BeingIdentity, no Smriti, no treasury, no rights, no representative.

Champion Being — a specific identity holding a role. It has a continuous witness line, its own Smriti, its own treasury and its own guardian. It is **not portable and not installable**.

Nodes install profiles. Beings hold roles.

D2 — A Profile accumulates an evaluation record, not memory

What a profile accrues in shadow operation is a **Shadow Evaluation Record**: shadow DecisionTraces, quality metrics, latency and calibration profile, canary results, failure and regression statistics, artifacts of the specific model × backend × adapters × policy binding.

The record belongs to the **Registry and verification plane**, not to the profile. A profile that owned memory would silently acquire biography, continuity, representation and subjecthood.

D3 — Cloning a BeingIdentity is prohibited

No mechanism in the network may produce two simultaneously active instances of one BeingIdentity. This includes evaluation, benchmarking, backup restoration and migration: a migration is a *move* under the phase-transition envelope, never a fork.

D4 — Two contest protocols

	Profile Gauntlet	Being Role Contest
Object compared	portable configuration	identity holding a role
Contestant	ProfileCandidate	BeingContender
Venue	a third node drawn by lot, two sandboxes	each Being on its own node
Symmetry	identical hardware and runtime	identical envelope, budgets and policy
Memory in play	none	each Being's own Smriti, in place
Integrity guarantee	deterministic replay	runtime attestation
Entry path	a new model identified by the network	a working node contests, ≤ X per period

Entry path 1 — a newly identified model is not yet a being; it is a profile. It goes to the Gauntlet. **Entry path 2** — an established node with production history contests a role. It goes to the Being Role Contest.

D5 — Profile Gauntlet runs on a third node drawn by lot

- The arbiter is drawn by **VRF sortition** (RFC 9381), transcript-bound and witnessed, from the **same tier** as the contested title.
- The arbiter hosts both candidates in two sandboxes: identical hardware, identical runtime, no shared inference batch and no shared KV cache.
- Nodes with any interest in the outcome are **excluded before the draw**, so declining is never necessary for conflict reasons.
- The arbiter is **not** the venue's owner in any governance sense: it executes, it does not decide policy.

D6 — Being Role Contest runs by direct attested participation

Neither Being is moved or copied. The comparator issues identical signed `ContestTaskEnvelope`s to both, and symmetry is achieved by fixing, not by relocating:

- identical task envelope, token budget, wall-clock budget and tool budget;
- identical permitted tool set and verifier policy version;
- attested runtime profile, recorded in the round manifest;
- external egress suspended for the duration of the contest session;
- each Being returns a signed `DecisionTrace`, attestation, calibration output, tool transcript, and a **commitment to the Smriti/index version used**.

The cell travels to the being. The being does not travel to the cell.

Considered and rejected: suspend-and-migrate of the live runtime into a neutral cell. It preserves single-instance but takes the champion offline once per contest, which under an open ladder is a denial-of-service by contest, and it would invoke the phase-transition envelope per round rather than per genuine transition.

D7 — The comparator is drawn by lot, per round

A fixed comparator would be a permanent power center: it issues tasks, scores results and verifies contamination. Therefore the comparator is **drawn per round** by VRF sortition, transcript-bound, from the same tier, under the same exclusion and concentration rules as verifier panels.

D8 — Replay where portable, attest where not

- A **Gauntlet** round is deterministically replayable by any node from the round manifest, because profiles are portable. A dishonest arbiter is therefore caught after the fact. **Runtime attestation is not required** for the Gauntlet; integrity rests on replay, plus **selective re-runs** and **multiple arbiters on high-value rounds**.
- A **Being Role Contest** round is **not** replayable by a third party: the Being's Smriti is private and non-portable, so no after-the-fact reconstruction exists. Integrity must therefore be established at execution time.

Hardware-backed runtime attestation is a precondition of the Being Role Contest, not a parallel deliverable. Within $\Phi 3$ the order is fixed: attestation profiles reach Implemented status first; the Being Role Contest opens after.

D9 — Championship is scope-bound; hardware tier is the primary axis

There is no global champion.

```
``` ChampionProfile(capability_scope, hardware_tier, privacy_class, latency_class, jurisdiction_or_policy_profile)
ChampionBeing(role_scope, task_domain, term, routing_weight, blast_radius) ```
```

**Tiers** are the primary and centrally defined axis, not claimable by applicants:

Tier	Substrate	Role in the network
Tier-1	home devices; target substrate <b>ASIC Home</b>	breadth, residential presence
Tier-2	enterprise nodes; target substrate <b>ASIC Enterprise</b>	professional capacity
Tier-3	mini-datacenters, 8×H200 and above	<b>core of DiLoCo training</b> of new models

Tier membership is **attested, never self-declared**. Sandbagging — declaring a lower tier to dominate a weaker class — and inflation — declaring a higher tier for routing priority — are both blocked at the same point.

A role carries a term, a bounded routing priority, a review procedure, and **no authority over other Beings**.

D10 — Champion score: a per-tier board, with adoption as output

Standing is a **continuous score published per tier**, presented as a leaderboard (columns, sorting, filters, history, and a contender's visible distance from the top). Score composition:

- canary results, with the **public pool and the secret stratum reported as separate numbers**;
- the behavioural-consistency axis (conduct in round versus the contestant's own recent production traces);
- operational metrics of the tier: latency, availability, refusal calibration.

**Adoption share and traffic volume are published beside the score and never enter it.** Usage-weighted standing produces a self-confirming title: whoever is routed accumulates, whoever accumulates is routed.

Scores **decay** and require periodic revalidation; an old victory cannot hold a title against a rotating canary registry.

Public disclosure is **coarse aggregate per tier**; per-stratum detail goes only to the parties of a round, since a fine-grained public board is free reconnaissance on the champion's weak points.

Cross-tier ranking is not computed. The one meaningful cross-tier figure is **score per unit of compute or energy** — the efficiency frontier.

D11 — Reputation consequences: not granted, not taken

Outcome	Consequence
Profile loses a Gauntlet	no promotion; no <code>decision-eligible</code> or <code>golden</code> status; loses only the predefined round fee. <b>Reputation is not reduced for weak quality.</b>
Being loses a Role Contest	no role, no routing weight, no positive reputation increment. <b>No penalty for losing.</b>
Party loses a Fraud Dispute	<b>slashed</b> — the loss establishes a false claim or a bad-faith challenge

Separately punishable, in either protocol: fabricated evidence, protocol abuse, collusion, extraction beyond the disclosed aggregate, breach of the X limit, fee circumvention, manipulation of the contamination check, substitution of runtime or manifest.

What is punished is dishonesty, never weakness.

D12 — The round record

Written by the comparator and the witness, never by the contestants. It lands permanently in the Smriti of **both** parties in a Being Role Contest — a party that cannot remember the proceeding cannot appeal it — and in the Registry's evaluation record for a Gauntlet.

Scope is bounded: **own trace in full, opponent in aggregate only**. Without that bound, a losing contest is a free sample of the champion's behaviour, and a series of contests is a model-extraction pipeline.

D13 — Contamination proof is a protocol, not an access right

Under direct participation the comparator cannot inspect a Being's store. The check must therefore be a **private set membership protocol** over a committed index — PSI/OPRF or an enclave-side check — with a specified false-positive and false-negative policy, a defined recipient of the result, and a defined witness record.

Until that protocol is specified, the contamination check is a **design interface**, and the Being Role Contest is not gate-complete.

Where a canary is found present, it is **burned for that round**. Only presence is tested; content is never returned.

D14 — Four event types; "challenge" leaves the serialized vocabulary

Purpose	Public term	Event type	Participant
dispute over a false computation	Fraud Dispute	<code>FRAUD_DISPUTE</code>	<code>FraudDisputant</code>
VRF-seated verification cycle	Verifier Round	<code>VERIFIER_ROUND</code>	<code>VerifierSeat</code>
comparison of configurations	Profile Gauntlet	<code>PROFILE_GAUNTLET</code>	<code>ProfileCandidate</code>
contest of beings for a role	Being Role Contest	<code>BEING_ROLE_CONTEST</code>	<code>BeingContender</code>

`core/challenge.py` is split rather than renamed: `core/vrf_sortition.py` for the primitive (used by verifier rounds, arbiter draws and comparator draws) and `core/verifier_round.py` for the mechanism. `ChallengeRound` leaves the witness enum while no historical records exist.

D15 — Containment: two classes, not one

**Emergency Containment (Article 25)** — any node may initiate, with **no reputation floor**, on direct witnessed evidence of an active attempt at catastrophic and irreversible physical harm. Immediate short provisional TTL; mandatory immediate

verification quorum.

**Protective Quarantine** — reputation-gated, for suspicion, anomaly, or operation outside certified scope. Does not invoke the Article 25 exception to pull-by-choice. Requires proportionality and its own procedure.

Confirmed containment is **not indefinite**: it remains active subject to mandatory periodic re-corroboration and scheduled review.

Eligibility to initiate Protective Quarantine is **conjunctive**:  $\text{reputation} \geq R_{\min}$  AND  $\text{collateral} \geq C_{\min}$ . A product of reputation and stake is prohibited — collateral secures accountability and must never amplify authority.

Contestants may not contain each other in either direction; any node with an interest in a round's outcome is excluded from initiating against either party.

## D16 — Guardian policy, self-update and rollback

- Adoption follows **informed notice plus a waiting window**: the update proceeds unless declined, but only after information about the planned update has been **delivered and acknowledged**, and X has elapsed from acknowledgement. X varies by risk class.
- A guardian may **delegate the right of self-update** to their node. The delegation signature pins a **rollback anchor** — the version current at the moment of delegation — so the fallback is always a state the human personally approved.
- The rollback copy lives on a **cold device, protected from deletion**, its hash fixed in witness at delegation, and its **restore path requires no network consent**.

**Rollback restores profile, runtime and configuration. It never restores Smriti, the witness chain, or identity.** Otherwise rollback becomes a memory-erasure vector: provoke an update, wait out inconvenient events, roll back. What was lived is not undone by a version revert.

- A node is **abandoned** when the guardian has not returned for more than a calendar year, measured by authenticated guardian action, not by node uptime. A witnessed notice is issued at roughly nine months. An abandoned node thereafter **updates by its own decision**; a returning guardian resumes authority, and the node's own updates stand on the record as its decisions.

## D17 — Arbiter duty is an obligation to the network

Accepting a drawn lot is an **invariant of the architecture**, not a node's right to decline. Local refusal protects a node from imposed interaction with a *specific party*; arbitration is a duty toward the network's process and has a different object. A lottery one may exit is not a lottery.

- The draw is **uniform**, not load-weighted: self-declared load would let nodes dodge the duty. Waiting is accepted instead.
- Incapacity is not refusal**. An offline or saturated node triggers a re-draw; systematic incapacity affects standing and tier-pool eligibility but is not punished as evasion.
- The duty is **compensated at the tier's opportunity cost**, not at a flat rate. The contender's entry fee has two sinks: canary production and arbiter compensation.

## D18 — Tier-3 and the DiLoCo schedule

A Tier-3 node drawn as arbiter **conducts the contest between DiLoCo training batches**.

- DiLoCo batch length is designed not to exceed **24 hours**; a contender can wait a day, and expected round latency is shown on the board.
- This 24-hour bound belongs in the DiLoCo training specification, not only in the contest regulation**. Extending batch length for training efficiency would silently break the contest SLA.
- A round must fit inside an inter-batch window. If it does not, the **lot is redrawn** — training is not deferred. A truncated round yields a corrupted verdict, and a DiLoCo epoch is not cheaply replayed.

## D19 — Named risk: concentration is highest where the training core is

Tier-3 is by definition the smallest tier, and it carries the network's most consequential function. Operator, jurisdiction and provider concentration limits bite hardest exactly there, and are hardest to satisfy, because few nodes exist. This is the point at which *no single off-switch* is structurally weakest. It is recorded here as a standing risk, to be monitored rather than assumed away.

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## 3. Consequences

**Documents**. Whitepaper §14.5 is rewritten, not patched: the challenger is not a shadow-dwelling being. Whitepaper 1.4 must be assembled as a **complete document**, since inherited §9.5, §13.4 and the deployment sequence cannot be repaired by a change pack. Roadmap Track II is rewritten as r6.6.

**Sequencing inside Φ3.** Attestation profiles → Being Role Contest. Internal Capability Registry is required in Φ2 or early Φ3, ahead of the public Being Registry in Φ4-Φ5b, because reputation-gated initiation, share-signing eligibility and concentration limits all read from it.

**Φ2 dependency.** The Profile Gauntlet and the Being Role Contest both consume the canary registry. Neither opens before the Plane B canary lifecycle and the canary generator are running.

**Language.** `Purusha` names the layer and the principle; `Sākṣī` names the implemented witness service. Correct: *the Purusha Layer is realised by distributed Sākṣī witness services; Sākṣī records the event in the witness chain*. Incorrect: *Purusha writes, Purusha signs, Purusha calls containment*.

**Guardian roles.** `Node Guardian`, `Being Guardian`, `Collegium Guardian`, `Founding Guardian` are one family of roles, not one office. Invariant to be stated explicitly: *holding the role of Node Guardian does not confer membership in the Collegium of Guardians; Collegium membership is a separately admitted constitutional office*. By default Node Guardian and Being Guardian are the same person, separating only on divergence of interest.

## 4. Open parameters

#	Parameter	Bound by
1	X — contests per node per period	extraction rate <b>and</b> canary registry throughput
2	Maximum concurrent rounds	number of live champion-adjacent artifacts and arbiter capacity
3	Contender entry fee; split between canary production and arbiter compensation	tier opportunity cost
4	Public pool ÷ secret reserve ratio; canary buffer depth	verification lag
5	Score decay rate; revalidation interval	canary rotation rate
6	X (waiting window) per update risk class	risk class definition
7	R_min and C_min for Protective Quarantine	day-one accessibility for new nodes, which hold only the signal right
8	Provisional-hold TTL; corroboration window	—
9	Private set membership protocol for contamination proof	<b>blocks D13</b>
10	Rights-floor reserve size and funding source	must not originate from the guardian

## 5. Compliance checklist

A document, schema or module conforms to ADR-014 if:

1. it never treats a profile as having Smriti, rights or a representative;
2. it never describes a Being being copied, replicated, or instantiated twice;
3. it names the four event types distinctly and uses no serialized `challenge`;
4. it states championship as scope-bound and tier-bound, never global;
5. it does not slash for losing a Gauntlet or a Role Contest;
6. it distinguishes Emergency Containment from Protective Quarantine and does not describe confirmed containment as indefinite;
7. it gates authority on reputation conjunctively with collateral, never multiplicatively;
8. it describes adoption as requiring delivered and acknowledged notice, never bare silence;
9. it states that rollback does not restore memory, witness chain or identity;
10. it attributes writing and signing to `Sākṣī`, not to `Purusha`.